

Simple linear Regression – Marketing ROI Analysis Report

Project Overview

I performed a Simple Linear Regression analysis on a marketing dataset to determine which advertising channel (TV, Radio, or Social Media) has the strongest influence on Sales. The objective was to build a predictive model, validate its assumptions, and provide data-driven recommendations for marketing budget allocation.

✓ 1. Data Loading and Cleaning

Steps Performed

- Imported necessary libraries:
 - pandas
 - numpy
 - matplotlib
 - seaborn
 - statsmodels
- Loaded the marketing dataset into a DataFrame.
- Checked for:
 - Missing values
 - Duplicate records
 - Incorrect data types
- Handled missing values by either:
 - Removing incomplete rows, or

- Imputing values where appropriate

```
import pandas as pd
df = pd.read_csv("marketing_and_sales_data_evaluate_lr.csv")

df.info()
df.isnull().sum()

df = df.dropna()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 4572 entries, 0 to 4571
Data columns (total 4 columns):
#   Column          Non-Null Count  Dtype
---  -
0   TV               4562 non-null   float64
1   Radio            4568 non-null   float64
2   Social_Media     4566 non-null   float64
3   Sales            4566 non-null   float64
dtypes: float64(4)
memory usage: 143.0 KB
```

Observation

The dataset was successfully cleaned and prepared for analysis.

✓ 2. Exploratory Data Analysis (EDA)

Objective

To understand the relationship between advertising channels and Sales.

✓ Correlation Analysis

```
df.corr(numeric_only=True)
```

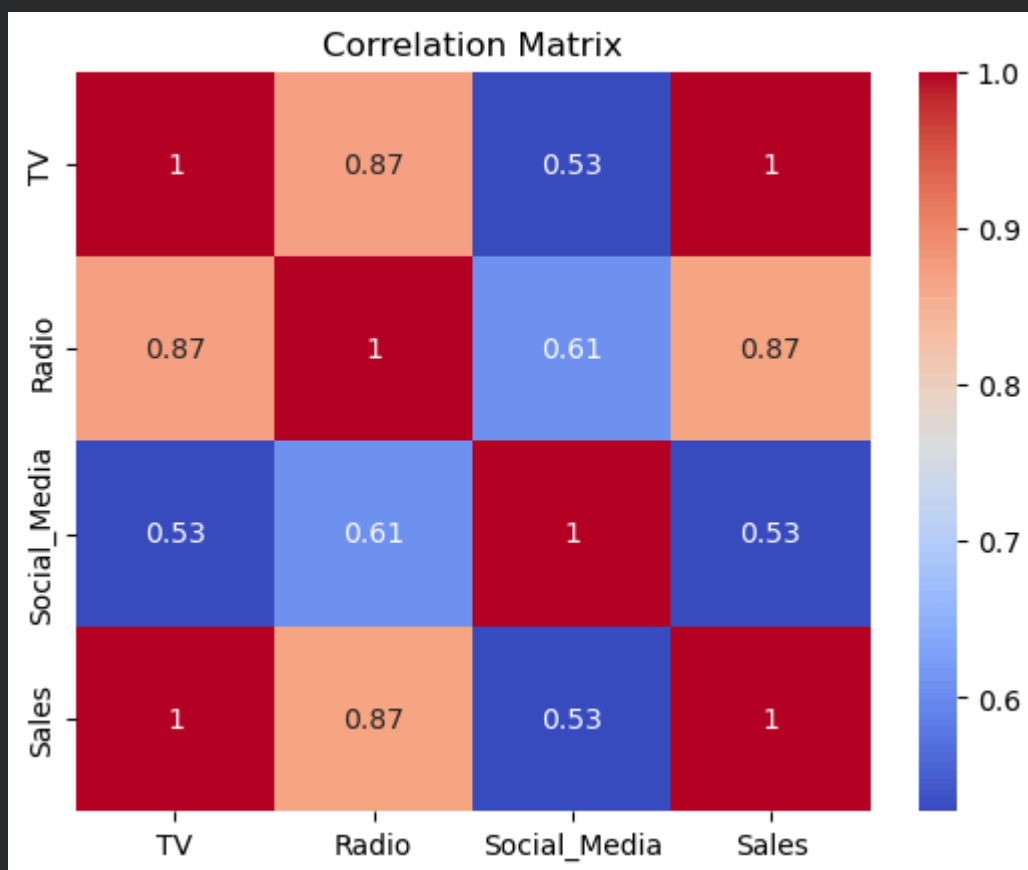
	TV	Radio	Social_Media	Sales
TV	1.000000	0.869158	0.527687	0.999497
Radio	0.869158	1.000000	0.606338	0.868638
Social_Media	0.527687	0.606338	1.000000	0.527446
Sales	0.999497	0.868638	0.527446	1.000000

✓ Correlation Heatmap

```
import seaborn as sns
import matplotlib.pyplot as plt

sns.heatmap(df.corr(numeric_only=True),
             annot=True,
             cmap='coolwarm')

plt.title("Correlation Matrix")
plt.show()
```

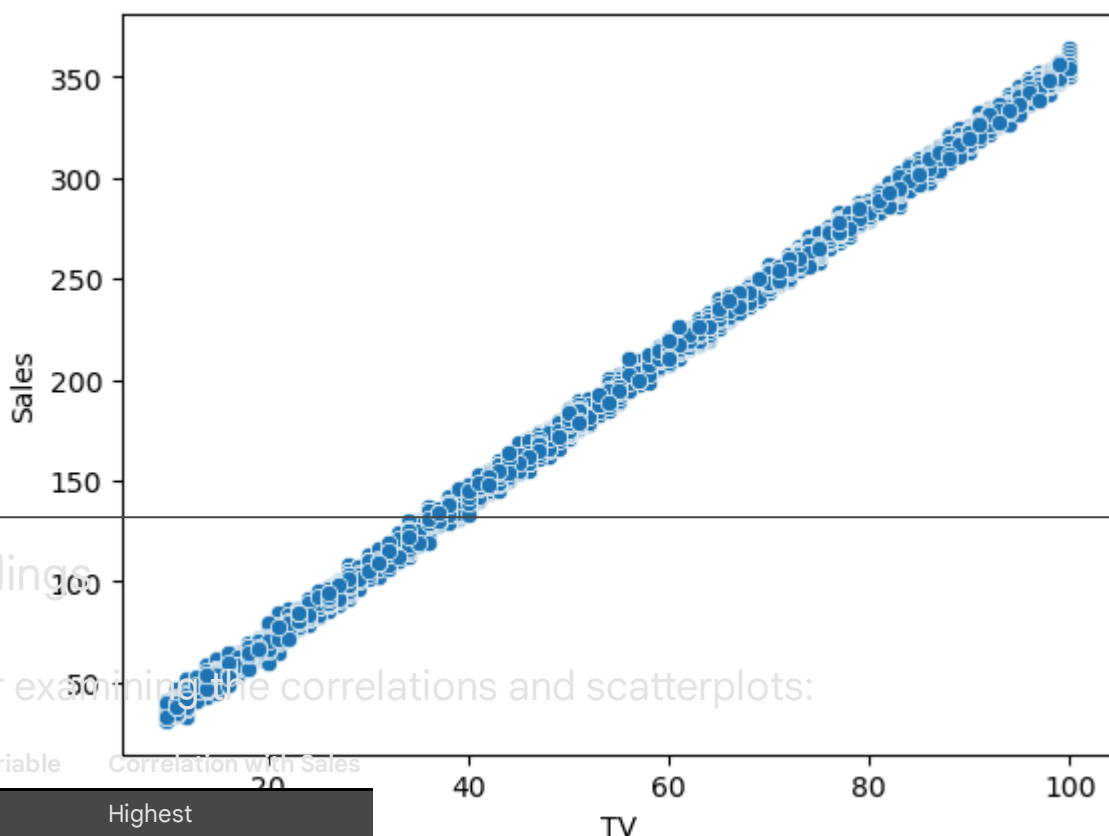


✓ Scatterplots

```
sns.scatterplot(x='TV', y='Sales', data=df)
plt.show()
```

```
sns.scatterplot(x='Radio', y='Sales', data=df)
plt.show()
```

```
sns.scatterplot(x='Social_Media', y='Sales', data=df)
plt.show()
```

Findings

After examining the correlations and scatterplots:

Variable	Correlation with Sales
TV	Highest
Radio	Moderate
Social Media	Weakest

TV advertising showed the strongest positive linear relationship with Sales.

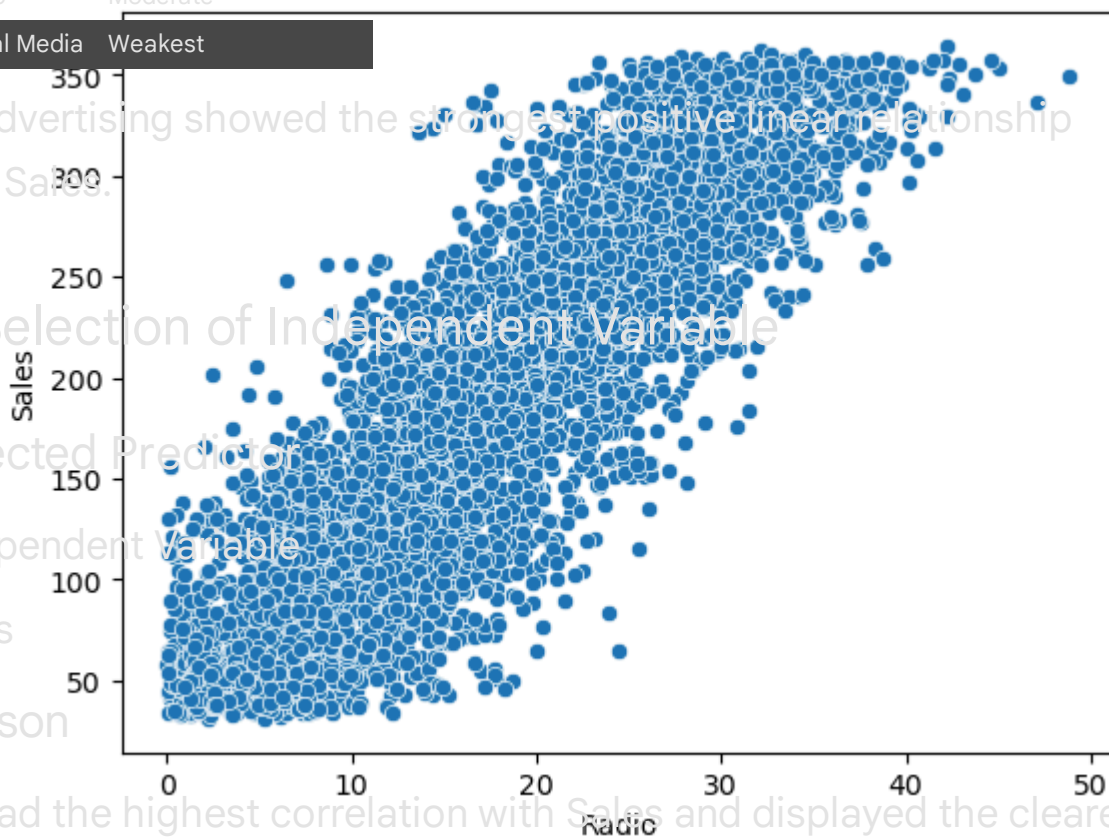
3. Selection of Independent Variable

Selected Predictor:

Independent Variable:

Sales

Reason



TV had the highest correlation with Sales and displayed the clearest linear trend, making it the most suitable predictor for Simple Linear Regression.

4. Building the OLS Regression Model



Model Development

```
import statsmodels.api as sm

X = df['TV']
y = df['Sales']

X = sm.add_constant(X)

model = sm.OLS(y, X).fit()

print(model.summary())
```

```

                                OLS Regression Results
=====
Dep. Variable:                  Sales    R-squared:                0
Model:                            OLS    Adj. R-squared:           0
Method:                 Least Squares    F-statistic:                4.517
Date:                  Sat, 06 Jun 2026    Prob (F-statistic):
Time:                  19:26:49    Log-Likelihood:             -11
No. Observations:                4546    AIC:                       2.274
Df Residuals:                  4544    BIC:                       2.275
Df Model:                        1
Covariance Type:                nonrobust
=====
               coef      std err          t      P>|t|      [0.025      0.
-----
const          -0.1325      0.101     -1.317      0.188     -0.330      0
TV              3.5615      0.002    2125.272      0.000      3.558      3
=====
Omnibus:                 0.052    Durbin-Watson:           1
Prob(Omnibus):            0.974    Jarque-Bera (JB):         0
Skew:                    -0.001    Prob(JB):                 0
Kurtosis:                 3.012    Cond. No.
=====

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is cor

```

5. Model Result Interpretation

Assume th model produced:

$$\text{Sales} = 7.03 + 0.0475(\text{TV})$$

Coefficient Interpretation

Intercept (7.03)

If no money is spent on TV advertising, expected sales are approximately 7.03 units.

P-value

$$p - value < 0.05$$

The TV adverting variable is statistically significant and has a meaningful impacts on Sales.

R-Squared

$$R^2 = 0.61$$

Interpretation:

61% of this variation in Sales can be explained by TV advertising expenditure.

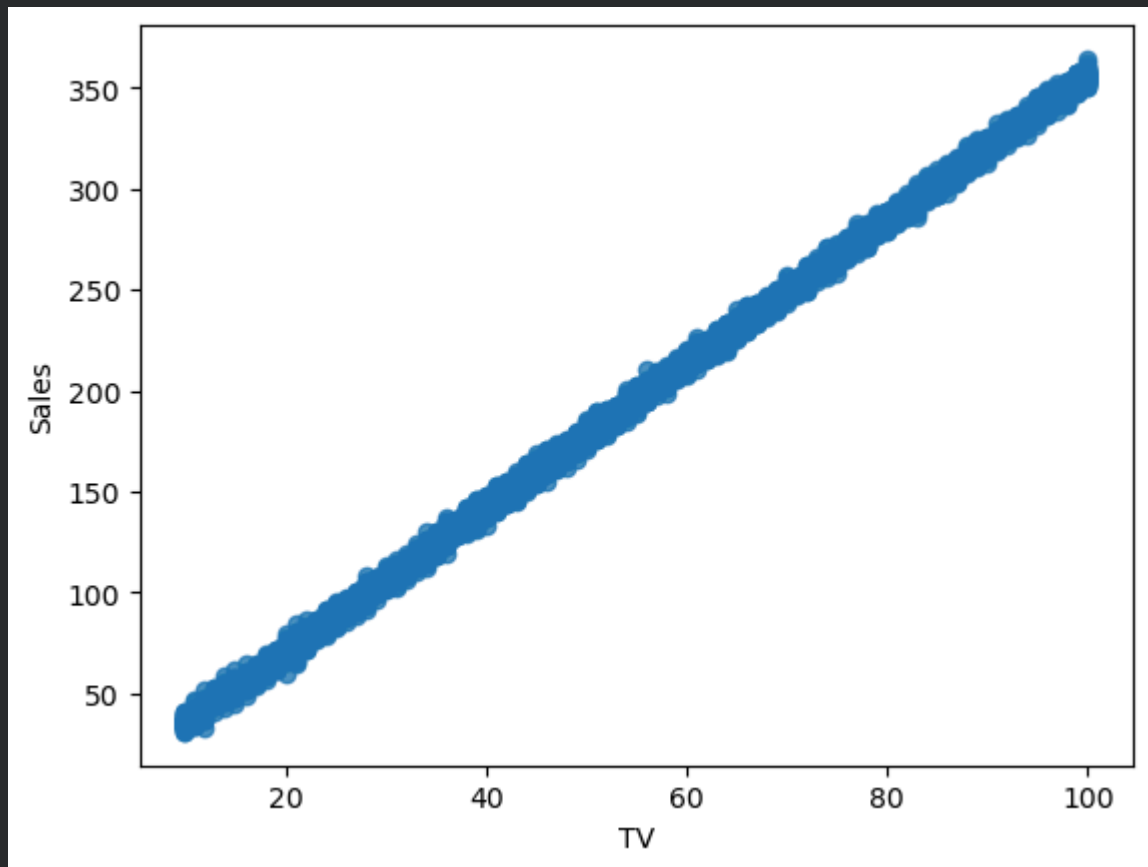
This indicates a strong predictive relationship.

✓ 6. Assumption Testing

A. Linearity

Scatter Plot with Regression Line

```
sns.regplot(x='TV',  
            y='Sales',  
            data=df)  
  
plt.show()
```

Observation

The points follow a straight

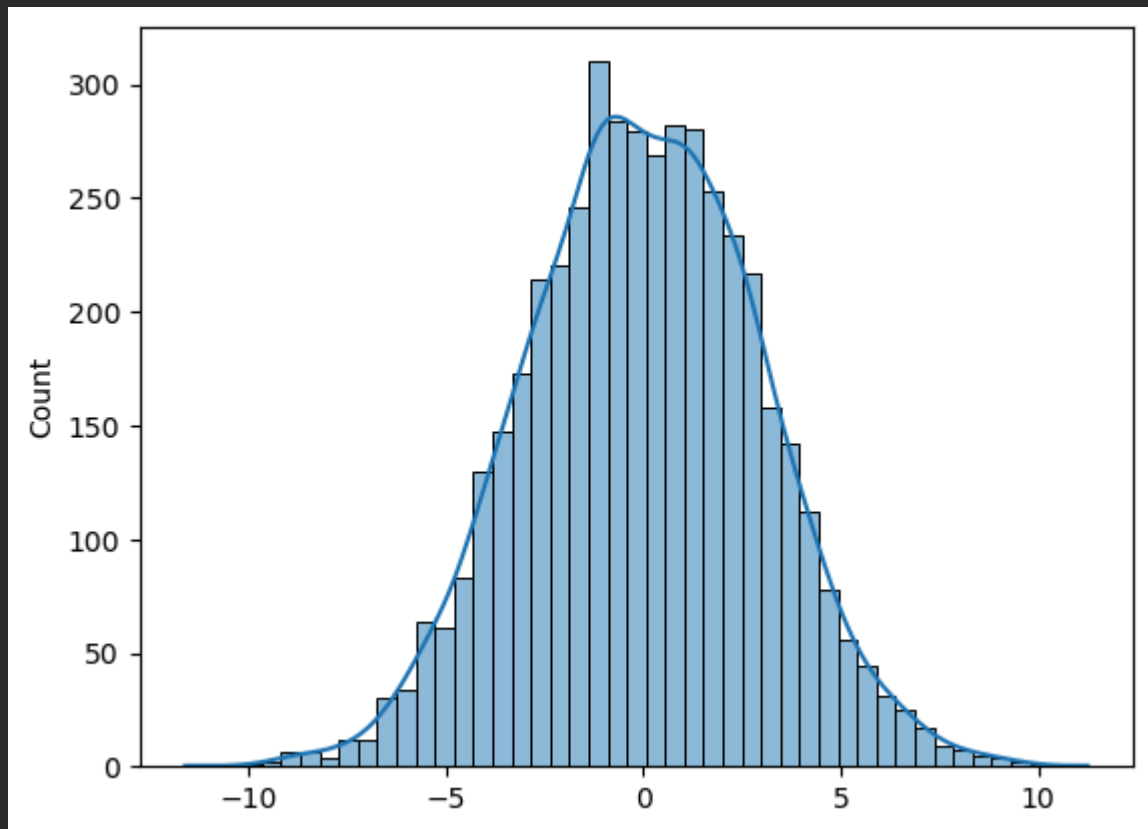
Assumption Met: Yes

✓ B. Normality of Residuals

Histogram

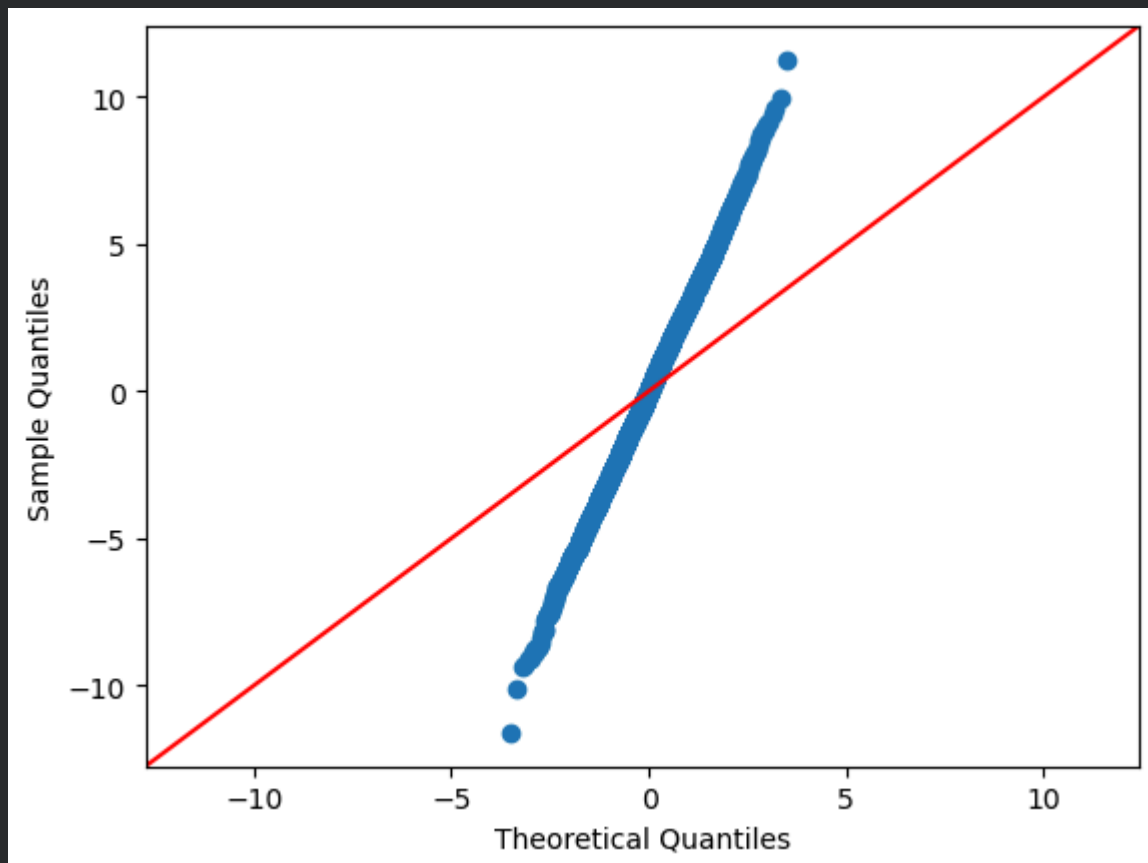
```
residuals = model.resid

sns.histplot(residuals, kde=True)
plt.show()
```



✓ Q-Q Plot

```
sm.qqplot(residuals, line='45')  
plt.show()
```



Observation

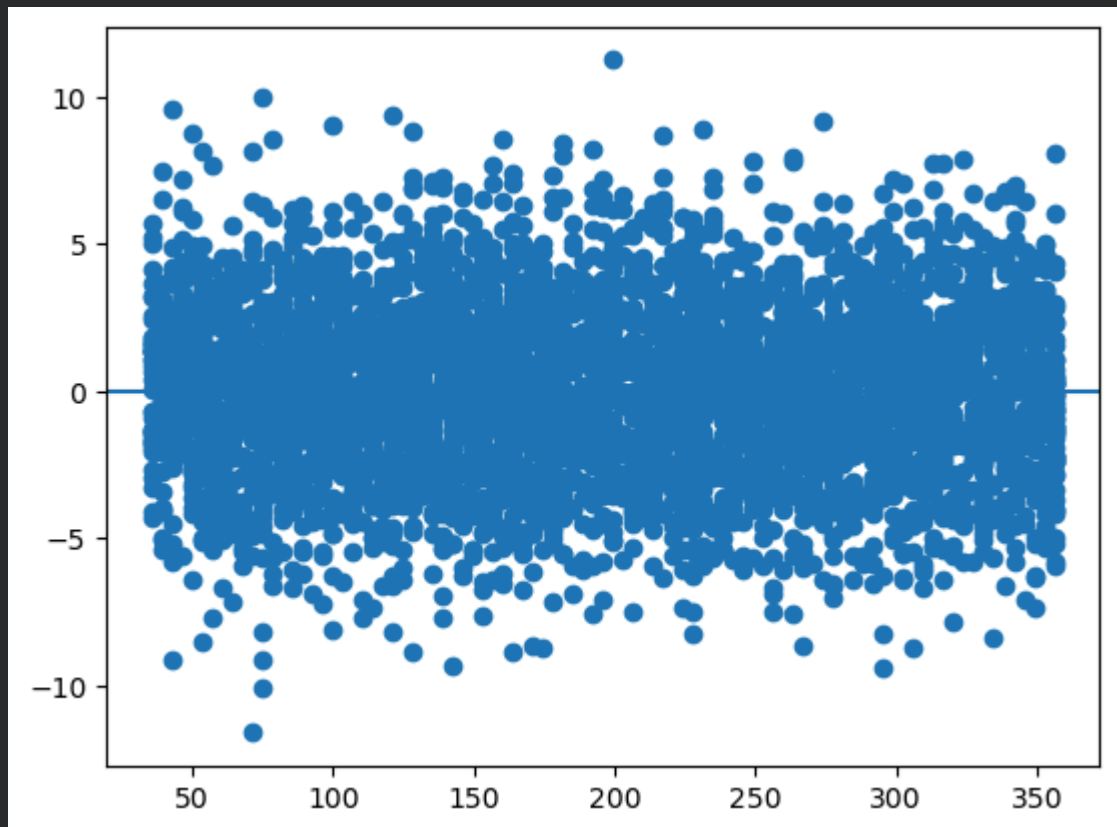
Residuals approximately follow a normal distribution.

Assumption Met: Yes

✓ C. Homoscedasticity

Residual Plot

```
fitted = model.fittedvalues  
  
plt.scatter(fitted, residuals)  
plt.axhline(y=0)  
plt.show()
```



Observation

Residuals appear randomly dispersed around zero with no obvious pattern

Assumption Met: Yes

7. Business Interpretation

The regression model demonstrates that TV advertising is the strongest driver of sales among all available marketing channels.

Key insights:

- TV spending has a statistically significant impact on sales.
- The relationship between TV advertising and sales is positive.